- 20260326

$$\frac{s+6}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3} = \frac{A(s+3)+Bs}{s(s+3)}$$

$$s+6 = A(s+3) + B.s$$

$$s=0 \longrightarrow 0+6 = A\cdot(0+3) + B.0 \longrightarrow A=2$$

$$s=-3 \longrightarrow -3+6 = A\cdot(-3+3) + B(-3) \longrightarrow B=-1$$

$$\frac{s+6}{s(s+3)} = \frac{2}{s} - \frac{1}{s+3} = Y(s)$$

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left(\frac{2}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+3}\right)$$

$$y(t) = 2 - e^{-3t}$$

- $$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Adj = \begin{bmatrix} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & +\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} \\ -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & +\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix} \\ +\begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 0 & 0 \end{vmatrix} & +\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$